Homework 1

In this set we fix a field extension $K \subset F$ unless specified otherwise.

1. (a) Show that $\alpha \in F$ is algebraic over K if and only if F contains a finite-dimensional K-vector *subspace* L such that $\alpha \cdot L \subset L$. (b) If the conditions of (a) are satisfied, let $A : L \to L$ be a K-linear operator of multiplication by α. Show that its minimal polynomial in the sense of linear algebra is equal to the minimal polynomial of α in field-theoretic sense.

2. Consider field extensions $E \subset F \subset K$. Suppose that F is algebraic over E and K is algebraic over F. Show that K is algebraic over E.

3. (a) Show that $f(x) = x^3+x^2+x+3$ is irreducible over $\mathbb{Q}$. (b) Consider the field $F = \mathbb{Q}(\alpha)$, where α is a root of $f(x)$. Express $\frac{1}{2-\alpha+\alpha^2}$ as a $\mathbb{Q}$-linear combination of 1, α, and α^2.

4. Find the degree (over $\mathbb{Q}$) of the splitting field of (a) $x^4+x^3+x^2+x+1$. (b) x^4-2.

5. (a) Suppose $[F : K] = 2$ and $\operatorname{char} K \neq 2$. Show that there exists $D \in K$ such that $F = K(\sqrt{D})$. (b) Show that (a) can fail if $[F : K] = \operatorname{char} K = 2$.

6. For all positive integers n and m, compute the degree $[\mathbb{Q}(\sqrt{n}, \sqrt{m}) : \mathbb{Q}]$.

7. Let $K \subset F$ be an algebraic extension and let R be a subring of F that contains K. Show that R is a field.

8. Let $f(x) \in K[x]$ be a polynomial of degree 3. Show that if $f(x)$ has a root in a field extension $K \subset F$ of degree 2 then $f(x)$ has a root in K.

9. Let $\alpha, \beta \in F$ be algebraic over K, let $f(x)$ and $g(x)$ be their minimal polynomials, and suppose that $\deg f$ and $\deg g$ are coprime. Prove that $f(x)$ is irreducible in $K(\beta)[x]$.

10. Let $K \supset \mathbb{Q}$ be a splitting field of a cubic polynomial $f(x) \in \mathbb{Q}[x]$. Show that if $[K : \mathbb{Q}] = 3$ then $f(x)$ has 3 real roots.

11. Let $F = K(\alpha)$ and suppose that $[F : K]$ is odd. Show that $F = K(\alpha^2)$.

12. Let $f(x) \in K[x]$ be an irreducible polynomial and let $g(x) \in K[x]$ be any non-constant polynomial. Let $p(x)$ be a non-constant polynomial that divided $f(g(x))$. Show that $\deg f$ divides $\deg p$.

13. Show that the polynomial $x^5 - t$ is irreducible over the field $\mathbb{C}(t)$ (here t is a variable). Describe a splitting field.

14. Let $\mathbb{F}_q$ be a finite field with q elements. Compute $\sum\limits_{a \in \mathbb{F}_q} a^k$ for $k > 0$.

15. (a) Show that the algebraic closure of $\mathbb{F}_p$ is the union of its finite subfields: $\bar{\mathbb{F}}_p = \bigcup\limits_{n=1}^{\infty} \mathbb{F}_{p^n}$. (b) Show that $\overline{\mathbb{F}}_p$ contains proper infinite subfields.

16. A complex number $\alpha \in \mathbb{C}$ is called *constructible* if $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is a power of 2. Assume without proof that α is constructible if and only if one can construct α (as a vector on the plane) using the ruler and the compass. (a) Show that $\cos 20^\circ$ is not constructible. (b) Show that trisection of an angle is not always possible using the ruler and the compass.

17. Let K_1, K_2 be subfields of a field K. Then *the composite field* K_1K_2 of K_1 and K_2 is the smallest subfield of K containing K_1 and K_2. Show that

$$[K_1K_2 : F] \leq [K_1 : F][K_2 : F],$$

with equality iff some basis of K_2 over F is linearly independent over K_1.